FRESHMEN ENGINEERING DEPARTMENT

I B Tech, I Sem

LINEAR ALGEBRA & CALCULUS (R-23)

(COMMON TO ALL BRANCHES)

QUESTION BANK (10M) 2023-2024

Module-1 MATRICES				
S.NO	QUESTIONS	CO	BL	Marks
1	(a) Obtain the rank of the following matrix, by reducing into the echelon form $\begin{bmatrix} 2 & -4 & 3 & -1 & 0 \\ 1 & -2 & -1 & -4 & 2 \\ 0 & 1 & -1 & 3 & 1 \\ 4 & -7 & 4 & -4 & 5 \end{bmatrix}$	1	2	5
	(b) Obtain the rank of the following matrix, by reducing into the echelon form $\begin{bmatrix} 2 & 1 & 3 & 5 \\ 4 & 2 & 1 & 3 \\ 8 & 4 & 7 & 13 \\ 8 & 4 & -3 & -1 \end{bmatrix}$	1	2	5
2	(a)Reduce the matrix $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$ to normal form and hence find the rank.	1	2	5
	(b)Reduce the matrix A= $\begin{bmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{bmatrix}$ to normal form and hence find the rank.	1	2	5

3	Find the non-singular matrices P & Q such that the normal form of A is PAQ, where A = $\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix}$. Hence find its rank.	1	1	10
4	Using the Gauss-Jordan method, find the inverse of the matrix $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$	1	1	10
5	Find the inverse of the matrix A using elementary operations (i.e., using Gauss-Jordan method). $A = \begin{bmatrix} -1 & -3 & 3 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 0 & 1 \end{bmatrix}$	1	1	10
6	(a) Solve the system of equations $x+y-3z+2w=0,\ 2x-y+2z-3w=0,\ -4x+y-3z+w=0,$ $3x-2y+z-4w=0.$ (b) Determine whether the following equations will have a non-trivial solution, if so solve them $4x+2y+z+3w=0,\ 6x+3y+4z+7w=0,\ 2x+y+w=0.$	1 1	3 5	5 5
7	Show that only real value of λ for which the following equations have non zero solution is 6 and solve them completely when $\lambda = 6$; $x+2y+3z=\lambda x,\ 3x+y+2z=\lambda y,\ 2x+3y+z=\lambda z.$	1	1	10
8	(a)Discuss for what values of λ and μ the equations x+y+z = 6, x+2y+3z=10, x+2y+λz = μ have (i) no solution (ii) a unique solution (iii)an infinite number of solutions . (b) Solve the system of equations by Gauss elimination method, if they are consistent $x+2y+z=3,\ 2x+3y+2z=5,\ 3x-5y+5z=2,\ 3x+9y-z=4.$	1 1	2 3	5 5
9	(a) Solve the system of equations by Gauss elimination method, if they are consistent $3x+3y+2z=1,\ x+2y=4,\ 10y+3z=-2,\ 2x-3y-z=5.$ (b) Solve the system of equations by Gauss elimination method, if they are consistent $5x+3y+7t=4,\ 3x+26y+2t=9,\ 7x+2y+10t=5.$	1 1	3 3	5 5
10	Solve the following system of equations by using Jacobi method (Correct to 3 decimal places): $8x-3y+2z=20,\ 4x+11y-z=33,\ 6x+3y+12z=35.$	1	1	10

11	Use Gauss-Seidal iteration method to solve the system $10x+y+z=12,\ 2x+10y+z=13,\ 2x+2y+10z=14.$	1	1	10
12	Solve the following system of equations by using Gauss-Seidal method (Correct to 3 decimal places): $8x-3y+2z=20,\ 4x+11y-z=33,\ 6x+3y+12z=35.$	1	1	10
Module-2 Eigenvalues, Eigenvectors and Orthogonal Transformation				
1	Find the Eigen values and Eigen vectors of $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$	2	1	10
2	Find the Eigen values and Eigen vectors of $\begin{bmatrix} 2 & 1 & -1 \\ 1 & 1 & -2 \\ -1 & -2 & 1 \end{bmatrix}$	2	1	10
3	Obtain the eigen values and eigen vectors of the matrix $A=\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$	2	2	10
4	Determine the eigenvalues and corresponding eigenvectors of the following systems: $10x_1+2x_2+x_3=\lambda x_1,\ 2x_1+10x_2+x_3=\lambda x_2,\ 2x_1+x_2+10x_3=\lambda x_3.$	2	5	10
5	Find a matrix P which transforms the matrix, $A=\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ to diagonal form.	2	1	10
6	Diagonalize the matrix A= $\begin{bmatrix} 2 & 1 & -1 \\ 1 & 1 & -2 \\ -1 & -2 & 1 \end{bmatrix}$	2	2	10
7	Verify Cayley-Hamilton theorem for the matrix A= $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$ and hence find its inverse.	2	2	10
8	Show that the matrix A = $\begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ satisfies its own characteristic equation and find A^{-1}	2	2	10

9	Find the characteristic equation of the matrix $A=\begin{bmatrix}1 & 1 & 3\\ 1 & 3 & -3\\ -2 & -4 & -4\end{bmatrix}$ and hence find its inverse. Use Cayley-Hamilton theorem.	2	1	10
10	State Cayley-Hamilton theorem and hence find its A^{-1}, A^{-2} if $A=\begin{bmatrix}1 & 2 & -2\\ 1 & -1 & 2\\ -1 & -4 & -3\end{bmatrix}$.	2	1	10
11	Find the Eigen vectors of the matrix $A=\begin{bmatrix}6 & -2 & 2\\ -2 & 3 & -1\\ 2 & -1 & 3\end{bmatrix}$ and hence reduce $6x^2+3y^2+3z^2-2yz+4zx-4xy$ to a sum of squares.	2	1	10
12	Reduce the real quadratic form $3x_1^2-3x_2^2-5x_3^2-2x_1x_2-6x_2x_3-6x_3x_1$ to canonical form.	2	1	10
	Module-3 **Mean Value Theorems(Calculus)**			
1	(a)Verify Rolle's theorem for $(x-a)^m(x-b)^n$ where m,n are +ve integers in [a,b]. (b)Verify Rolle's theorem for the function log $[\frac{x^2+ab}{x(a+b)}]$ in [a, b] , a >0, b >0.	3 3	2 2	5 5
2	(a)Verify Rolle's Theorem for the function f(x)= $\frac{\sin x}{e^x}$ in $[0,\pi]$ (b) Verify Rolle's Theorem for $f(x)=x(x+3)e^{-x/2}$ in the interval $[-3,0]$	3 3	2 2	5 5
3	(a)Write Geometrical interpretation of Rolle's Theorem (b)Verify Lagrange's Mean value Theorem for the function f(x) = (x-1)(x-2)(x-3) in the interval [0, 4]	3 3	2 2	5 5

4	Using Lagrange's Mean value theorem Show that $\frac{v-u}{1+v^2} < \tan^{-1} v - \tan^{-1} u < \frac{v-u}{1+u^2}$, $0<u<v$ and hence deduce the following (i) $\frac{\pi}{4}+\frac{3}{25} < \tan^{-1}\left(\frac{4}{3}\right) < \frac{\pi}{4}+\frac{1}{6}$ (ii) $\frac{5\pi+4}{20} < tan^{-1}2 < \frac{\pi+2}{4}$	3	2	10
5	(a) Verify cauchy's Mean Value theorem for the functions e^x and e^{-x} in [a, b]. (b) Verify cauchy's Mean Value theorem for the functions f(x) =$\sqrt{x}$ and g(x) = $\frac{1}{\sqrt{x}}$ in [a, b].	3 3	2 2	5 5
6	(a)Verify Taylor's theorem for f(x) = (1- x)$^{\frac{5}{2}}$ in [0,1] with Lagrange's form of remainder upto 2 terms. (b) Show that log(1+x) = x-$\frac{x^2}{2}$ + $\frac{x^3}{3}$ - $\frac{x^4}{4}$ +.....	3 3	2 2	5 5
7	(a)Calculate the approximate value of $\sqrt{10}$ to Four decimal Places using Taylor's theorem. (b) Expand f(x)= Tanx up to the 5th powers of 'x' by Maclaurin's series and hence find series for log (sec x).	3 3	2 2	5 5
8	(a) Find the Taylor's series expansion of sin2x about x=$\frac{\pi}{4}$. (b)Show that log(1+ e^x) = log2 + $\frac{x}{2}$ + $\frac{x^2}{8}$ - $\frac{x^4}{192}$ + ---------- and hence deduce that $\frac{e^x}{e^x+1}$ = $\frac{1}{2}$ + $\frac{x}{4}$ - $\frac{x^3}{48}$ + ------	3 3	1 2	5 5
9	(a) Verify Rolle's theorem for f(x) = $\lvert x \rvert$ in [-1,1]. (b) Verify Rolle's theorem for f(x) = $e^x(sinx - cosx)$ in [$\frac{\pi}{4}$, $\frac{5\pi}{4}$].	3 3	2 2	5 5
10	(b)Verify Lagrange's Mean value Theorem for the function f(x) = x^3-x^2-5x+3 in the interval [0, 4] (b)Verify Lagrange's Mean value Theorem for the function f(x) = Cosx in the interval in [0,$\frac{\pi}{2}$].	3 3	2 2	5 5
11	(a) Expand log $_e$x in powers of (x – 1) and hence evaluate Log 1.1 correct to 4 decimal places. (b)Show that $\frac{sin^{-1}x}{\sqrt{1-x^2}}$ = x+$\frac{4x^3}{3!}$+ ---------	3 3	2 2	5 5

12	Prove that $\frac{\pi}{6}+\frac{1}{5\sqrt{3}} < sin^{-1}\left(\frac{3}{5}\right) < \frac{\pi}{6}+\frac{1}{8}$.	3	2	10
Module-4 **Partial differentiation and Applications (Multi variable calculus)**				
1	(a) Verify $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$ for the function u = $\tan^{-1}(\frac{x}{y})$. (b) If $u = x^2 + y^2 + z^2 : x = e^t; y = e^t sint;$ $z = e^t cost;\ then\ find\ \frac{du}{dt}$.	4 4	2 2	5 5
2	If u = f(r,s,t) where r = $\frac{x}{y}$, s = $\frac{y}{z}$ and t = $\frac{z}{x}$ show that x $\frac{\partial u}{\partial x}$ + y $\frac{\partial u}{\partial y}$ + z $\frac{\partial u}{\partial z}$ = 0.	4	2	10
3	Expand sin(xy) in powers of (x-1) and (y- $\pi/2$) upto second degree terms.	4	2	10
4	(a) If $u = \frac{yz}{x}$, $v = \frac{zx}{y}$, $w = \frac{xy}{z}$ show that $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ = 4 (b) If u= x + y+ z, uv = y + z, uvw=z, then prove that $\frac{\partial(x,y,z)}{\partial(u,v,w)}$ = u^2 v.	4 4	2 2	5 5
5	Verify Whether the following functions are functionally dependent, if so, find the relation between them U = $\frac{x+y}{1-xy}$, V= $\tan^{-1}x + \tan^{-1}y$.	4	2	10
6	Investigate the maxima and minima, if any of the function f(x,y) = x^3y^2(1-x-y)	4	3	10
7	(a)A rectangular box open at the top has volume of 32 cubic feet. Find the dimensions of the box requiring least material for its construction. (b)Find a point on the plane 3x+2y+z-12 = 0 which is nearest to the origin	4 4	2 2	5 5
8	Given that x+y+z = a, find the maximum value of $x^m y^n z^p$	4	2	10
9	Expand e^xsiny in powers of x and y	4	2	10
10	(a)If u=e^{x+y} , v=e^{-x+y} then find jacobian. (b)Prove that $u = x + y + z,\ v = xy + yz + zx, w = x^2 + y^2 + z^2$ are functionally dependent and find the relation between them.	4 4	2 2	5 5
11	(a) Find the minimum value of $x^2+y^2+z^2$ given that xyz = a^3 (b)Find the minimum value of $x^2 + y^2 + z^2$, given that $ax + by + cz = P$	4 4	2 2	5 5

12	Examine for maxima and minima of sinx +siny +sin(x+y)	4	2	10
	Module-5 **Multiple Integrals (Multi variable Calculus)**			
1	(a) Evaluate $\int_0^1 \int_x^{\sqrt{x}} x^2y^2(x+y)dydx$ (b) Evaluate $\int_0^1 \int_0^{\sqrt{(1+x^2)}} \frac{dydx}{1+x^2+y^2}$	5 5	2 2	5 5
2	Evaluate $\iint_R y\, dxdy$where R is the region bounded by the parabolas y^2 =4x and x^2=4y	5	2	10
3	Evaluate$\iint r \sin\theta\, drd\theta$ over the cardioids r =a (1-cos θ) above the initial line.	5	2	10
4	Evaluate$\int_0^a \int_0^{\sqrt{a^2-x^2}} \sqrt{x^2+y^2}\, dxdy$ by changing into polar coordinates.	5	2	10
5	Change the order of integration and evaluate$\int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dydx.$	5	2	10
6	(a) Find the area of the region bounded by the parabolas $y^2 = 4ax\ \ and\ x^2 = 4ay$ (b) Find the area of the loop of the curve r= a (1+cos θ)	5 5	2 2	5 5
7	Evaluate $\int_1^e \int_1^{logy} \int_1^{e^x} logz\, dzdx\, dy$	5	2	10
8	Using double integration , Find the volume of the tetrahedron bounded by the coordinate planes and $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}$=1.	5	2	10
9	Find by triple integral the volume of the tetrahedron bounded by the planes x=0, y=0, z=0 and $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}$=1.	5	2	10
10	Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} y^2 dxdy$ by changing the order of integration.	5	2	10
11	(a) Evaluate$\int_0^5 \int_0^{x^2} x(x^2+y^2)dxdy$ (b) Evaluate $\iint_R (x^2+y^2)\, dxdy$,where R is the square $0\leq x \leq a, 0 \leq y \leq a$.	5 5	2 2	5 5
12	Find $\iiint_R (x+y+z)dzdydx$ where R is the region bounded by the planes x=o, x=1,y=0, y=1, z=0, z=1.	5	2	10